

1. Write a program in C to check if a given number is even or odd using the function.
2. Write a C program to find the maximum and minimum of three numbers.
3. Write a program in C to check whether a number is a prime number or not using the function.
4. A positive integer is entered through the keyboard. Write a function to obtain the prime factors of this number. For example, prime factors of 24 are 2, 2, 2 and 3, whereas prime factors of 35 are 5 and 7.
5. Write a function to calculate the sum of digits of a number. Firstly, there will be a prompt asking the user the number of digits.
6. Write a function power (a, b), to calculate the value of a raised to b. (Making your own pow function)
7. Write a program in C to check Armstrong and **Strong (a number that equals the sum of the factorials of its digits)** numbers using the function.
8. Evaluate the following series using functions. Name your user-defined function as seriesEvaluator.

$$1+2+3+\dots +N$$

$$1*2*3*4*\dots *N$$

$$\frac{1}{2^3} + \frac{1}{4^3} + \frac{1}{6^3} + \dots + \frac{1}{N^3}$$

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{N}$$

$$\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots + \frac{1}{N!}$$

$$\frac{1*2}{3!} + \frac{2*3}{4!} + \frac{3*4}{5!} + \dots$$

9. Given three variables x, y, z write a function to circularly shift their values to right. In other words if **x = 5**, **y = 8**, **z = 10** after circular shift **y = 5**, **z = 8**, **x = 10**. Call the function with variables **a, b, c** to circularly shift values.
10. Write a program in C to print the first 50 natural numbers using recursion. (similar to Hello World printing)
11. Write a program in C to find the sum of the first 50 natural numbers using recursion.
12. Write a program in C to find the product of the first 50 natural numbers using recursion.